RENTALSHIELD

Evidence-Based Rental Listing Screening

Project Documentation

Datathon Project | Vancouver, British Columbia, Canada

Group SPARKLERS_14

Yamini Sairam - Sairam.y@northeastern.edu

Gayathri Sairam - sairam.g@northeastern.edu

Gayathri Srinivasan - srinivasan.gaya@northeastern.edu

.

Document Control

Item	Description
Project name	RentalShield
Document type	Technical and product documentation
Project category	Civic technology / renter decision support / applied data analytics
Pilot geography	Downtown Vancouver and West End, British Columbia
Primary users	Renters reviewing an online rental listing before paying or sharing sensitive information
Application pattern	Flask web application with data-processing and verification modules
Version	1.0 — Datathon demonstration documentation
Status	Working prototype; findings and score are indicative, not validated predictions

Executive Summary

RentalShield is a renter-focused screening prototype designed to make fragmented rental-verification work easier to understand. Before committing money to an online rental listing, a renter often needs to check whether the address appears in public property records, whether the advertised rent differs materially from a market benchmark, what police-reported incidents have been recorded near the location, and whether the listing description or copied landlord chat contains known warning phrases.

The application combines these checks into one explainable report. The current pilot focuses on Downtown Vancouver and the West End. It uses City of Vancouver property-address data, Statistics Canada rent benchmark extracts, Vancouver Police Department crime data, and transparent rule-based phrase matching. The report presents an overall concern score, component results, a concise crime-context summary, recommended verification actions, and—when the score is elevated—alternative neighbourhoods to explore.

RentalShield is a decision-support tool, not a fraud classifier trained on confirmed scam cases. Its score is a provisional rule-based heuristic, not a probability that a listing is fraudulent. A property-address match does not verify ownership or the advertiser's authority; low rent is not proof of fraud; and nearby crime records describe the surrounding area rather than the legitimacy of an individual rental listing.

Table of Contents

NO	CONTENT
1	Problem Definition and Motivation
2	Project Aim, Objectives and Scope
3	Users and Use Cases
4	System Architecture and Workflow
5	Data Sources and Preparation
6	Verification Methodology
7	Scoring and Decision Logic
8	Alternative-Neighbourhood Recommendations
9	User Interface and Report Design
10	Technology Stack and Project Structure
11	Setup and Execution
12	Validation and Testing Strategy
13	Responsible Use, Privacy and Limitations
14	Evaluation Plan and Success Metrics
15	Future Enhancements
16	Conclusion
17	References

1. Problem Definition and Motivation

1.1 Problem statement

Online rental listings can be difficult to assess before a viewing or payment. Relevant evidence is distributed across property records, rental-market statistics, police-reported incident datasets, the wording of the listing, and guidance from public authorities. Renters may need to navigate these sources independently and may not know how to interpret a missing record, a large price difference, or a request for an advance deposit.

RentalShield addresses this information-fragmentation problem by bringing several evidence checks into a single workflow and explaining what each result does—and does not—mean.

1.2 Why this problem matters

- **Financial exposure**: an advance payment can be difficult to recover if the listing is deceptive.
- **Information asymmetry**: the advertiser may know more about the property and their authority to rent it than the prospective tenant.
- **Time and cognitive burden**: renters may have to search multiple websites and interpret unfamiliar public datasets.
- **False confidence**: a real address, plausible rent, or absence of obvious warning phrases can be mistaken for proof of legitimacy.

1.3 Product proposition

> **"Found a rental? Check the evidence before you pay."**

RentalShield is not a rental marketplace and does not certify landlords. It acts as an evidence layer that supports more informed verification decisions.

2. Project Aim, Objectives and Scope

2.1 Aim

To build an explainable web-based prototype that consolidates public-data checks and listing-language signals to help renters assess online rental listings before making a payment.

2.2 Objectives

- Verify whether a submitted address matches the available property-address dataset for the supported pilot areas.
- Compare the entered monthly rent with a suitable Vancouver Census Metropolitan Area (CMA) rent benchmark for the selected bedroom category.
- Summarize police-reported incident types and relative incident activity within approximately 500 metres of a matched address, using available 2023–2026 data.
- Detect configured warning phrases in a listing description or pasted landlord/agent chat.
- Combine the component results into a transparent, provisional concern score and explain the factors that contributed to it.
- Offer actionable verification steps and, for elevated scores, suggest alternative neighbourhoods to investigate using the available address and crime comparison data.
- Communicate data limitations so that users do not mistake screening indicators for definitive fraud or safety determinations.

2.3 Scope

In scope for the pilot	Outside the current verified scope
Downtown Vancouver and West End address checks	All Canadian cities or all Vancouver neighbourhoods for address verification
Address presence in the supplied public dataset	Proof of title, ownership, landlord identity, or authorization
Vancouver CMA rent benchmark comparison	A unit-specific appraisal or guaranteed neighbourhood rent estimate
Approximate nearby police-reported incident context	Official population-adjusted crime rate or guarantee of personal safety
Rule-based warning-phrase detection	A trained, validated machine-learning fraud probability model
Neighbourhoods to explore as alternatives	Confirmation that a specific rental unit is currently available at the entered price

3. Users and Use Cases

3.1 Primary user

A renter who has found a listing on a rental platform, classifieds site, or social-media group and wants to complete a quick evidence review before sending money or sensitive information.

3.2 Main user journey

- Open RentalShield and enter the civic number, street, supported area, monthly rent, bedroom count, and listing description or copied conversation.
- Submit the form for verification.
- Review the address match result and any relevant matched address record.
- Review the rent difference against the Vancouver CMA benchmark.
- Read a concise nearby-crime summary based on the matched address coordinates and available incident records.
- Review detected warning phrases and their explanations.
- Review the overall concern score, its component breakdown, and personalized actions.
- If the score is Medium or High, use suggested neighbourhoods as starting points for further rental searches.
- Download or revisit the report if the application exposes those options for the saved report.

3.3 Core use cases

Use case	Input	Expected output
Address presence check	Civic number, street, area	Found, not found, invalid input, or unsupported-area result with a clear explanation
Rent comparison	Monthly rent and bedroom category	Benchmark amount, percentage difference, and interpretation
Nearby crime context	Matched address coordinates and area	Common recorded incident types and relative activity comparison over available years
Text screening	Listing description or pasted chat	Configured warning types, severity, matched phrases, and caution message
Overall assessment	Results from available checks	Provisional score, component breakdown, explanation, and suggested actions
Alternative-area discovery	Entered rent and current area context	Neighbourhoods to explore, not verified available properties

4. System Architecture and Workflow

4.1 Logical architecture

Layer	Responsibilities	Representative components
Presentation	Collect listing inputs and present the report	Flask templates, HTML, CSS, JavaScript, Leaflet map
Application	Route requests, validate inputs, manage user/report workflows	app.py and Flask routes
Verification and scoring	Run individual checks and combine outputs	address_checker.py, rent_checker.py, listing_analyzer.py, crime_checker.py, risk_engine.py
Data access	Load and normalize CSV-based public datasets	data_loader.py and baseline-building scripts
Persistence	Store users and saved report data where configured	SQLite database and application data-access code
External/public evidence	Supply property, rent and crime context	City of Vancouver, Statistics Canada, Vancouver Police Department

4.2 Processing workflow

User submits listing details

↓

Input validation and address normalization

↓

Address lookup ─ Rent benchmark ─ Crime context ─ Text warning rules

↓

Component results + evidence explanations

↓

Provisional concern score and recommended actions

↓

Report page and saved/exportable report (where configured)

4.3 Design principles

- **Modularity**: each check has a focused responsibility and can be tested independently.
- **Explainability**: the report should show which evidence or phrase caused a warning.
- **Graceful degradation**: when a dataset or check is unavailable, other results should remain visible where possible.
- **Separation of evidence types**: address presence, price difference, local incident context, and text warnings are not interchangeable facts.
- **Conservative claims**: outputs should describe evidence and uncertainty rather than label a person or property as fraudulent.

5. Data Sources and Preparation

5.1 City of Vancouver Property Addresses

Purpose: determine whether the entered address can be matched to an available public property-address record and, where present, obtain the geographic coordinates needed for the nearby-crime lookup.

Pilot extracts are stored separately for Downtown and West End, while a citywide property-address extract is used to build citywide reference locations for crime comparison. Address normalization accounts for common street suffixes, punctuation, spacing, and directional prefixes.

Interpretation: a match confirms presence in the searched dataset only. It does not prove that a unit exists at the advertised configuration, that it is available, or that the advertiser owns or is authorized to rent it. A non-match can result from incomplete data or formatting differences and is not proof of fraud.

5.2 Statistics Canada rent benchmark

Purpose: compare the entered monthly rent with a published Vancouver CMA benchmark for the chosen bedroom category. The project uses Table 46-10-0092-01 and local extracts for one-bedroom, two-bedroom, and three-or-more-bedroom units.

Bedroom category	Illustrative Q2 2026 asking-rent benchmark used in the project
1 bedroom	CAD $2,280/month
2 bedrooms	CAD $3,030/month
3 or more bedrooms	CAD $4,150/month

Percentage difference is calculated as: (advertised rent − benchmark rent) ÷ benchmark rent × 100. A negative result means the advertised rent is below the benchmark; a positive result means it is above the benchmark.

These figures are CMA-level benchmarks, not street-level or property-specific valuations. Benchmark data should be refreshed when newer releases become available, and the report should display the data period used.

5.3 Vancouver Police Department crime data

Purpose: provide nearby reported-incident context around the submitted address. The project combines incident extracts for 2023, 2024, 2025, and 2026 year-to-date where available. The data includes incident type, year, neighbourhood and projected coordinates.

The nearby search uses an approximate 500-metre radius around the matched property coordinates. The citywide comparison baseline is built from sampled citywide property-address reference points for the same year and radius. The report should summarize common recorded incident types and one combined-period relative comparison rather than presenting a separate percentage for every year.

Important interpretation: the source records represent police-reported incident records, not a population-adjusted crime rate. Incident coordinates can be generalized for privacy. Some records may be duplicate-looking because the source extract does not provide a unique incident identifier; therefore counts should be described as records, not guaranteed unique crimes.

5.4 Listing and landlord-chat text

Purpose: identify predefined phrases that may indicate pressure to pay, a request for a deposit before viewing, inability to show the property, or unusual payment methods. The current method is deterministic rule-based pattern matching using regular expressions.

The result should include the warning category, severity, matched phrase, and plain-language explanation. This is not a full semantic understanding system and may miss indirect wording or flag benign phrases that require context.

5.5 Data inventory and lineage

Dataset / artifact	Role in RentalShield	Key caveat
property_addresses_downtown.csv	Address match for Downtown pilot	Not a complete registry of units or rental availability
property_addresses_west_end.csv	Address match for West End pilot	Same limitation; format normalization required
vancouver_property_addresses.csv	Citywide reference points for baseline building	Used for comparison sampling, not unit availability
rent_benchmark_1br_q2_2026.csv	One-bedroom rent comparison	Vancouver CMA level
rent_benchmark_2br_q2_2026.csv	Two-bedroom rent comparison	Vancouver CMA level
rent_benchmark_3br_q2_2026.csv	Three-or-more-bedroom comparison	Vancouver CMA level
crimedata_csv_AllNeighbourhoods_AllYears.csv	Incident-level citywide crime context	Year-to-date 2026; coordinate and record-count caveats
crime_reference_counts_500m_<year>.csv	Citywide comparable-area baseline by year	Depends on reference-point and preprocessing methodology

6. Verification Methodology

6.1 Address verification

The user-provided civic number and street are normalized and compared with the supported area's property-address data. The check should distinguish at least four outcomes: found, not found, invalid input, and unsupported area. A technical failure to load or process the dataset should not be silently reported as an ordinary non-match.

6.2 Rent benchmark comparison

The rent checker selects the benchmark corresponding to the bedroom category and computes the percentage difference. The output should show the advertised rent, benchmark rent, bedroom category and signed percentage difference. The concern score uses the documented threshold rules; the numerical difference remains visible so users can interpret the result.

6.3 Nearby crime context

The crime checker uses the matched address coordinates to select records within the configured radius. It summarizes frequent incident types and compares the address area's count with comparable sampled 500-metre reference locations for the selected years. For a concise user experience, the interface combines the period into a short summary, while the underlying data retains the year-level values for traceability.

A percentile is a relative ranking among the sampled comparison locations. For example, a high percentile means the location's reported-incident count is higher than most sampled comparison points under the selected method; it does not mean that the same percentage of all crimes in Vancouver occurred there.

6.4 Listing-language screening

The analyzer searches the supplied text for configured patterns and returns warning types, severities and matched phrases. Example categories include payment within a short deadline, deposit-to-secure wording, deposit before viewing, inability to show the property, and unusual payment methods. Warnings are evidence for further review, not definitive classifications.

6.5 Evidence trail

Each component should preserve the input used, status, source or comparison period, key output values, and explanation. This allows a renter, reviewer or judge to understand why a result appeared and supports reproducibility during a demo.

7. Scoring and Decision Logic

The product's demonstration design allocates up to 25 points to each of four checks, for a maximum overall concern score of 100. The score is a transparent, provisional heuristic intended to prioritize follow-up—not a trained or calibrated probability of fraud.

Component	Maximum	Scoring interpretation
Address verification	25	An address not found in the searched dataset can contribute concern points; a match contributes 0 address concern points. Invalid input or technical failure should be shown as incomplete rather than treated as a confirmed non-match.
Nearby crime activity	25	Relative activity labels are mapped to a bounded contribution (for example, Low = 0, Medium = 13, High = 25) for the product's chosen demo rules. This describes area context, not listing fraud.
Rent benchmark difference	25	Larger configured rent anomalies receive more points. The signed percentage and benchmark must remain visible; above-benchmark rent should not automatically be described as scam evidence.
Listing/chat warning wording	25	Configured phrase warnings receive points according to severity, with the component capped at 25.
Total	100	Sum of available component points, with missing checks clearly identified.

7.1 Score interpretation

Score band	Label	Recommended interpretation
0–24	Low concern	Fewer configured warning indicators were detected. This does not establish legitimacy.
25–49	Medium concern	Pause and complete additional verification before paying.
50–100	High concern	Several or strong indicators warrant careful independent checks before proceeding.

7.2 Missing or unavailable evidence

The system should not silently assign a reassuring result when a check could not run. The report should distinguish 'not found' from 'could not be completed'. If one or more components are unavailable, it should label the assessment as partial and disclose which checks were not completed. This is especially important for unsupported areas, missing coordinates, absent user text, or missing benchmark data.

7.3 Avoiding misleading score claims

- Do not call the score a fraud probability, fraud rate, or official safety rating.
- Do not imply that high nearby crime proves a rental listing is fraudulent.
- Do not imply that a matching address verifies ownership, landlord authority or availability.
- Do not interpret no detected text warnings as proof that the listing is legitimate.
- Treat the score thresholds and component weights as provisional until evaluated against reviewed examples and user testing.

8. Alternative-Neighbourhood Recommendations

8.1 Purpose

When the overall concern score is Medium or High, RentalShield is designed to help the renter continue their search by suggesting up to three alternative neighbourhoods to investigate. The monthly rent entered for the listing is reused as the renter's search target; a separate budget field is not required for the current workflow.

8.2 Intended selection logic

- Use the submitted listing's supported area as the starting context.
- Aggregate comparable reported-incident counts by neighbourhood using consistent years and methodology where the available data supports it.
- Identify neighbourhoods with lower relative reported-incident activity than the current area, without describing them as guaranteed safe.
- Use the citywide property-address dataset to ensure that suggested areas have address records and to support geographic plausibility.
- Display neighbourhood names and the entered monthly rent as a search target, not as verified local rent or proof of available units.
- Encourage the renter to check current listings and compare their actual price, bedroom count, lease terms and location independently.

8.3 Data limitation

The available property-address dataset does not contain current rental availability, unit bedroom count, or confirmed asking rent for each property. Statistics Canada's benchmark is Vancouver CMA-wide. Therefore, recommendations should be phrased as 'areas to explore' rather than 'available apartments at this rent'. If neighbourhood-specific rental listings or rent statistics are integrated later, the system can improve budget matching.

8.4 Recommendation transparency

For each suggested area, the interface should explain why it was selected (for example, lower relative incident activity in the comparison period and presence of property-address records), state the comparison period, and avoid presenting a small difference as a meaningful safety advantage without validation.

9. User Interface and Report Design

The interface uses a dark navy visual theme with teal and amber accents. The homepage introduces the problem and collects listing details. The report page prioritizes a total score and four compact component summaries, then provides concise crime context, recommendations, limitations, and report actions.

9.1 Report design requirements

- Show the overall score and the component contribution of each of the four checks.
- Use a short crime summary: common incident types, one relative citywide comparison, and the crime component contribution.
- Avoid repeating the same explanatory text in every alternative-neighbourhood card.
- Make found/not-found/unavailable states visually distinct and use plain language.
- Keep the difference between area context and listing-specific signals explicit.
- Provide PDF/JSON exports only where the corresponding route and saved-report data are available.

9.2 Maps

The homepage includes separate map panels for Downtown Vancouver and the West End. The current map implementation uses Leaflet with CARTO basemap tiles and OpenStreetMap attribution. Leaflet itself does not require an API key, but the selected tile provider may require a key or impose usage conditions. The key should be configured through an environment variable and restricted according to the provider's guidance.

9.3 Accessibility and clarity

- Use descriptive headings, readable contrast and plain-language explanations.
- Do not rely on color alone to communicate Low, Medium or High status.
- Keep error messages actionable and distinguish user-input errors from data-source failures.
- Avoid excessive annual crime cards when a concise combined-period summary is more useful.

10. Technology Stack and Project Structure

Technology / component	Role
Python	Application logic, data processing and scoring
Flask	Web server, routes, form handling and report rendering
Pandas / NumPy	CSV ingestion, normalization and aggregation
SQLite (where configured)	Local persistence for application records and saved reports
HTML / CSS / JavaScript	User interface and interactive report elements
Jinja2	Server-side HTML templating
Leaflet + CARTO tiles	Interactive area maps
Regular expressions	Rule-based warning-phrase matching
CSV files	Prepared public datasets and comparison baselines

10.1 Key files

File / directory	Responsibility
app.py	Flask routes, form handling, authentication/report workflow and template rendering
data_loader.py	Loads and validates public CSV datasets
address_checker.py	Normalizes and matches property addresses
rent_checker.py	Computes rent benchmark differences
listing_analyzer.py	Detects configured listing/chat warning phrases
crime_checker.py	Computes nearby incident summaries and comparison levels
build_citywide_crime_baseline.py	Builds citywide reference counts for comparable 500 m locations
risk_engine.py	Orchestrates checks, calculates the score, builds explanations and recommended actions
templates/index.html	Homepage, rental-check form and map panels
templates/analytics.html	Analytics dashboard
templates/report_detail.html	Saved report and score breakdown
data/	Public datasets and processed comparison files
requirements.txt	Python dependencies
.env	Local secrets and environment-specific configuration; must not be committed

11. Setup and Execution

11.1 Prerequisites

- Python installed on the development machine.
- A virtual environment for project dependencies.
- The project's required CSV datasets in the expected data folders.
- Environment variables configured locally where needed.

11.2 Installation

From the project root directory, activate the virtual environment and install dependencies:

```
pip install -r requirements.txt
```

11.3 Prepare the crime baseline

If the project uses the citywide reference baseline, run the baseline-building script after the source crime CSV and citywide property-address reference file are in their expected locations:

```
python build_citywide_crime_baseline.py
```

11.4 Run the application

Start the Flask application using the command configured by the project. In a typical local setup:

```
python app.py
```

Open the local address printed by Flask in a browser. Use a fresh verification submission after code or scoring changes; older saved reports may preserve results generated by the earlier code version.

11.5 Environment and secret management

Keep secret values in a local .env file or environment variables, never directly in templates or committed source files. Ensure .env is listed in .gitignore. A basemap API key exposed to browser code should be treated as a public client key and restricted by allowed origins, usage limits and provider controls.

11.6 Dataset path checks

- Confirm filenames and folder names match the loaders exactly.
- Check delimiters; the property-address extracts may use semicolons.
- Check coordinate columns and their order before distance calculations.
- Confirm the baseline files were regenerated after changes to the reference data or calculation method.
- Do not treat a successful file load as proof that all records are clean or complete.

12. Validation and Testing Strategy

12.1 Unit tests

Module	Representative test cases
Address checker	Known matching address; non-matching address; invalid civic number; unsupported area; punctuation and suffix normalization
Rent checker	Below benchmark; equal to benchmark; above benchmark; invalid rent; unsupported bedroom category; missing benchmark value
Listing analyzer	Deposit before viewing; payment deadline; unusual payment method; benign text; empty input; multiple phrases; repeated phrase
Crime checker	Known coordinates; no records in radius; missing coordinates; unsupported area; partial year; absent baseline file; duplicate-looking rows
Risk engine	All checks complete; one or more checks unavailable; score caps; correct component sum; clear explanation; recommendations generated only when appropriate
Report template	All check statuses render; long text wraps; no duplicated sections; missing values do not break page; score and breakdown agree

12.2 Integration tests

- Submit a complete listing and confirm that all four component results appear in the report.
- Confirm the displayed total equals the sum of component points and never exceeds 100.
- Confirm an address-not-found result is distinct from an address-verification technical error.
- Confirm the crime summary uses the expected radius and data period and does not show an unsupported population-adjusted rate.
- Confirm Medium/High results trigger alternative-area suggestions when sufficient comparison data exists.
- Confirm missing alternative data produces a clear unavailable state rather than fabricated recommendations.
- Confirm report HTML, JSON export and PDF export (where implemented) display consistent results.

12.3 Datathon demo test scenario

Use a controlled, clearly labelled demonstration listing with a known address, a rent value for a selected bedroom category, and text containing one or more configured warning phrases. Verify the outputs before recording. Do not describe test values as live listings or real fraud cases. Keep a screenshot or run log showing the input, each component output, total score, and recommended action.

12.4 Evaluation metrics

- Functional correctness: percentage of test cases producing expected outputs.
- Score consistency: every report total matches its component breakdown.
- Coverage: percentage of submitted test scenarios for which each check can return a valid or explicit unavailable status.
- Explainability: a reviewer can identify why each point contribution was made.
- Usability: a first-time user can complete a check and explain the result without developer assistance.
- Recommendation validity: suggested areas meet documented comparison criteria and are not represented as available rentals.
- Robustness: malformed input or missing data does not crash the whole report.

13. Responsible Use, Privacy and Limitations

13.1 Not a fraud determination

RentalShield should not state or imply that a particular landlord, property manager, or property is fraudulent based solely on the score. It should report observable indicators and recommend independent verification.

13.2 Public data limitations

- Public address datasets can omit valid addresses, be delayed, or use a different naming convention.
- An address match does not prove ownership, authority to rent, unit availability or the existence of the advertised unit configuration.
- The rent benchmark is CMA-level and may not reflect building quality, furnished status, utilities, lease duration or other unit-specific factors.
- Police-reported incidents are not the same as population-adjusted crime rates, and the location data may be generalized.
- 2026 is a partial year in the current dataset and should not be compared with full years without a clear qualification.
- Incident records may include duplicate-looking rows when no unique incident identifier is available.
- Rule-based text screening cannot reliably interpret every phrase, language, context or evasion strategy.
- Alternative neighbourhood recommendations are area suggestions, not confirmed current rental listings.

13.3 Privacy and security

- Collect only the listing information required for screening.
- Avoid requesting identity documents, bank details or unnecessary personal data.
- Protect authentication credentials and report records using appropriate access controls.
- Keep secrets out of source control and use environment configuration.
- Do not expose private user-submitted chats or identifying information in public demos.

13.4 Fairness and unintended effects

Neighbourhood-level crime indicators can be misunderstood or stigmatize communities. RentalShield should describe the data and comparison method, avoid labeling neighbourhoods as 'safe' or 'unsafe', and never use area-level incident counts as a proxy for an individual landlord's honesty. User testing should check whether the report encourages proportionate, evidence-based actions rather than panic.

13.5 Appropriate user-facing wording

Avoid saying	Prefer saying
This rental is a scam.	The available evidence includes warning indicators; verify independently before paying.
This address is safe because it was found.	The address matches the available property-address dataset; ownership and authority were not verified.
The area has a crime rate of X% based on incident counts.	Reported incident activity is higher than X% of comparable sampled locations, under this method.
No warning phrases means the listing is legitimate.	No configured warning phrases were detected in the supplied text.
These apartments are available at your budget.	These neighbourhoods may be worth exploring; current availability and local rents were not verified.

14. Evaluation Plan and Success Metrics

A datathon prototype should be evaluated not only by whether it runs, but by whether it gives users a clear, defensible and useful result. The following evaluation plan is appropriate for the current stage.

Evaluation dimension	Method	Evidence to present to judges
Problem relevance	Map features to documented renter decisions and public scam guidance	One-minute user journey and clearly scoped problem statement
Data provenance	Record source, period, fields, transformations and caveats	Dataset inventory and reproducible preparation steps
Technical reliability	Run repeatable unit and integration tests	Test matrix and successful sample run
Explainability	Trace each point to a rule and input evidence	Score breakdown and matched-phrase explanation
User experience	Observe first-time users completing a check	Time-to-completion and confusion points
Recommendation quality	Manually verify the area-ranking output against stored baseline values	Example alternative-area output with transparent criteria
Responsible use	Review language for overclaiming and stigmatizing interpretations	Limitations section and careful interface copy

For a stronger future evaluation, collect a small, consented and reviewed set of example listings, label each warning signal with human review, compare rule outputs with those labels, and report precision/recall only when the label quality and sample size support meaningful estimates. Do not claim model accuracy for a rules engine without such evaluation.

15. Future Enhancements

- Add a reviewed rental-listing dataset or permitted platform integrations to support real, current availability checks.
- Integrate neighbourhood-specific rent data or verified listing prices to improve budget-matched alternative recommendations.
- Add an optional link-validation module with clear limits on what URL checks can establish.
- Expand address verification beyond the current two-neighbourhood pilot after validating data coverage and matching quality.
- Improve multilingual and context-aware text analysis, retaining matched evidence and human-readable explanations.
- Build automated regression tests for scoring, report rendering, baseline calculations and exports.
- Track source refresh dates and display data freshness on each report.
- Conduct usability testing with renters and revise wording based on comprehension rather than visual appeal alone.
- Only consider supervised machine learning after a sufficiently large, lawfully collected, reviewed and representative labelled dataset exists.

16. Conclusion

RentalShield demonstrates how public data and transparent rules can be assembled into a practical renter-support workflow. Its contribution is not a claim that a score can definitively identify a scam. Instead, it reduces the effort required to review several evidence sources, explains uncertainty, and gives the renter concrete next steps before paying.

The most defensible path to a strong datathon submission is to demonstrate a complete, repeatable journey: a real user problem, traceable data, correct and explainable checks, a readable report, responsible interpretation, and recommendations that do not claim more than the data can support. Continued work should prioritize validation, data freshness, recommendation quality and user comprehension.

References

1. British Columbia RCMP. Rental Scams. https://rcmp.ca/en/bc/safety-tips/frauds-and-scams/rental-scams
2. Richmond RCMP. Rental Deposit Scams (public warning). https://rcmp.ca/en/bc/richmond/news/2025/08/richmond-rcmp-warns-public-about-rental-deposit-scams
3. Statistics Canada. Table 46-10-0092-01: Asking and paid rent prices, by geography and unit characteristics. https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=4610009201
4. City of Vancouver Open Data. Property Addresses. https://opendata.vancouver.ca/explore/dataset/property-addresses/
5. City of Vancouver Open Data. Rental Standards: Current Issues. https://opendata.vancouver.ca/explore/dataset/rental-standards-current-issues/
6. Competition Bureau Canada. How to report fraud and scams in Canada. https://competition-bureau.canada.ca/en/fraud-and-scams/tips-and-advice/how-report-fraud-and-scams-canada